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INITIAL SYSTEM PROPOSAL

FOR Hasta Travel Vehicle Rental Management System

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1.0 Introduction

In today's era characterized by technological innovation, mobility, and convenience, technological systems play a rather important role in reshaping conventional service sectors. Hasta Travel is a reliable mobility partner for students and staff of Universiti Teknologi Malaysia (UTM). It offers vehicle rental services to enable convenient movement in and around campus. However, with a growing customer base, there are rather important limitations to conventional processes.

At present, Hasta Travel relies on WhatsApp to handle its reservation and booking process and uses WhatsApp to communicate directly with customers. Customers must first contact a staff member for vehicle availability. Staff then respond manually using pre-set messages with a booking form supported by Whatsapp. Customers must fill in their personal details, such as name, matric or staff ID, phone number, and intended rental dates. Staff manually verify vehicle availability once the form is completed and contact information, check the eligibility of the customer, and confirm that payment has been processed. Hence, because of using only WhatsApp, it will delay in response times, cause overlapping bookings, highly depend on staff availability as well as increase the risk of data inconsistency and human error. From a data management perspective, the absence of a centralized database system makes it difficult to track rental histories, monitor fleet usage, manage financial records, and ensure accountability across transactions.

To address these problems, this project seeks to develop a comprehensive Vehicle Rental Management System for Hasta Travel. The proposed system aims to digitalize and streamline the vehicle rental process for UTM students and staff. Through this system, customers can easily make bookings and view vehicle availability in real time, while administrators can manage user data, vehicle information, bookings, payments and blacklisted accounts. The system will ensure efficient user verification, secure payment handling, transparent booking management and a reward mechanism in order to encourage customer loyalty. This truly enhances the convenience, trust and also operational efficiency.

Overall, this project proposal represents a strategic initiative that aims to upgrade Hasta Travel's operations by replacing manual workflow systems with a centralized, and scalable vehicle rental management system. This project will clearly define the system boundaries, user roles, and data flows that could eventually help in the service quality improve.

2.0 Background Study

2.1 Organisation Background

Hasta Travel is an incorporated private limited company Sdn. Bhd. that operating within Universiti Teknologi Malaysia (UTM). It provides vehicle rental services to UTM staff and students and was established in 2017 with a focus on low-cost vehicle rental services at the UTM Student Mall. In 2020, the company obtained its licence as a travel and tour operator.

Currently, it serves over 200 students per month, handling a significant amount of booking, deposit, and renting data. Being a sales-oriented business, all its operations largely depend on manual management and simple accounting software. All data, from bookings to customer and payment data, is dispersed through WhatsApp messages, WhatsApp forms, Excel, and paper documents.

With the implementation of compulsory LHDN e-Invoicing requirements starting June 2025, the company's manual methods of operation appear to face the data inaccuracies, and compliance issues for its operations as well as the efficiency of operations. An integrated, centralised information system is required for Hasta Travel to support its continued growth in operations and the enhancement of service delivery quality.

2.2 Case study of the existing system

Today, the vehicle rental management process at Hasta Travel is operated through semi-digitization and the manual system. Most core business activities such as booking management, customer verification, payment confirmation, and record keeping require extensive manual intervention by staff, leading to operational inefficiencies and a higher risk of human error.

Customers use WhatsApp to begin the booking process by contacting Hasta Travel to complete a pre-set WhatsApp form. Then, customers must wait for confirmation from the employee before they will be able to view available vehicles and confirmation from the booking process. To do this, the employee will have to go into the internal employee admin management system called Driveo to find out what vehicles are available and how many vehicles are in the fleet. However, because the employee has no way to use this internal employee admin management system, they will have to send customers messages through WhatsApp with the details of the vehicle rental booking and payment information, if available.

Currently, most Hasta Travel's payments are received from customers via QR codes submitted through WhatsApp to be verified by Hasta Travel's staff. There is a lack of integration for payments, deposits, and bookings, which leads to difficulties in refunds and management of outstanding payments. Inspections for returning vehicles are carried out in a traditional manner, where records are not integrated for vehicle inspections and customer rental records. Customer documents are also managed in a separate system for effective verification processes.

From an accounting perspective reconciliation of deposits, rents, and refunds is done manually using spreadsheets. It is inefficient, prone to errors, and not adequate for meeting the requirements of LHDN e-Invoice. Because of these concerns, Hasta Travel plans to transform to a fully integrated vehicle rental management system that allows for central management of all data, automated management of processes, and increased efficiencies with operations.

3.0 Requirement Analysis (based on AS-IS analysis)

The current vehicle rental operation is semi-manual and relies heavily on WhatsApp, Google Drives, Excel files and staff coordination. Most processes are executed manually following a detailed Standard Operating Procedure (SOP).

3.0.1 As-Is Booking Process

In the current practice, customers make vehicle rental bookings through WhatsApp. Staff will verify the vehicle availability using the whiteboard booking schedule and the Wahdah system. The customer eligibility which is limited to UTM students or staff is checked manually by staff. Proof of deposit payment is also submitted by customers via WhatsApp. Subsequently, staff are required to download deposit receipts, organize and store them in monthly folders, and manually enter booking details into multiple systems. All booking-related activities are carried out manually by staff, with no system automation involved.

This process is highly dependent on staff accuracy and attentiveness, which increases the probability of human error. Checking vehicle availability manually across multiple platforms will introduce a risk of double booking, particularly during peak periods. In addition, the absence of a real-time automated availability system will prevent the instant booking confirmation and cause delays in responding to customers. Furthermore, manual verification of customer eligibility and payment validation increases operational workload and slows down processing time. As a result, any miscommunication or oversight by staff may lead to booking inconsistencies, missing records, and customer dissatisfaction.

3.0.2 As-Is Vehicle Pickup Process

In the current practice, staff manually update the booking status in the Wahdah system before generating the required rental documents. These documents are then downloaded, printed, and sent to customers via WhatsApp. Customers are required to sign the documents manually and submit photographs of the vehicle condition through WhatsApp. After that, staff print and file the signed agreement form, inspection form, and customer-related documents for record-keeping purposes.

This process results in excessive printing and a heavy reliance on physical paperwork, which increases administrative workload and operational costs. The repeated handling of printed documents raises the risk of document mismatch, loss, or misfiling, especially when managing multiple rentals simultaneously. In addition, the absence of a centralized digital document and signature system makes it difficult to track document versions, verify authenticity, and retrieve records efficiently.

3.0.3 As-Is Return Process

In the current practice, vehicle return time is tracked using a GPS tracking system, namely Tracksolid Pro. Staff are required to manually review and adjust the return time, also record the vehicle's mileage, fuel level, and any damages upon return. Charges for late returns, fuel shortages, and rental extensions are calculated manually and recorded in Excel. Communication regarding deposit refunds is handled through WhatsApp.

This process reveals several issues. Manual calculations significantly increase the risk of errors, particularly when multiple vehicles are returned within a short time frame. The use of multiple disconnected systems, including GPS tracking software, Excel spreadsheets, and WhatsApp, leads to complicated data management and increases the probability of inconsistencies and miscommunication. In addition, the absence of an automatic penalty computation mechanism means that deductions rely entirely on staff judgment and accuracy, which may result in calculation discrepancies, delayed processing, and disputes with customers.

3.1 Current business process (scenarios workflow)

3.1.1 Car Booking Workflow

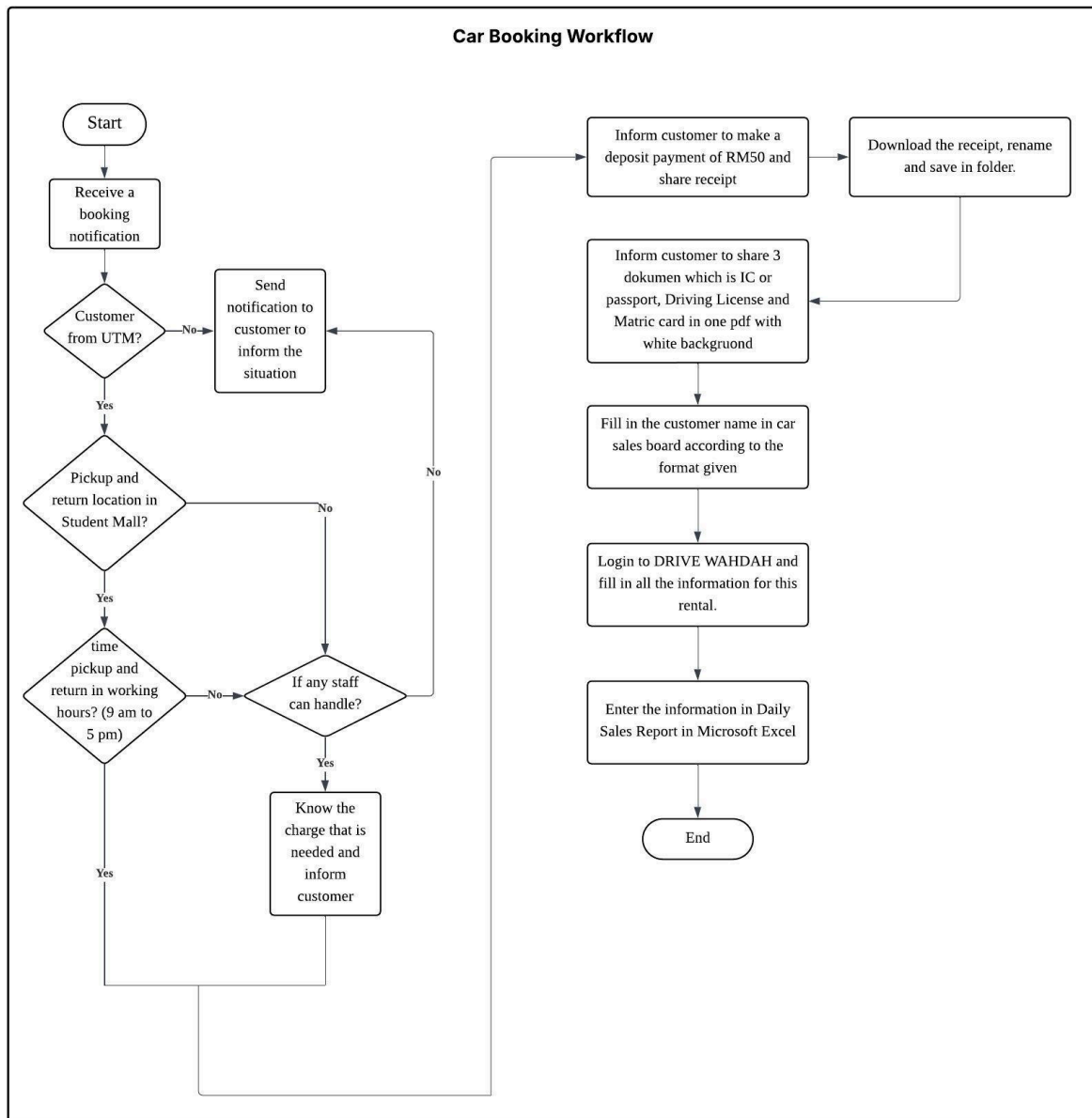


Figure 3.1.1.1 Car Booking Workflow

3.1.2 Car Pickup Workflow

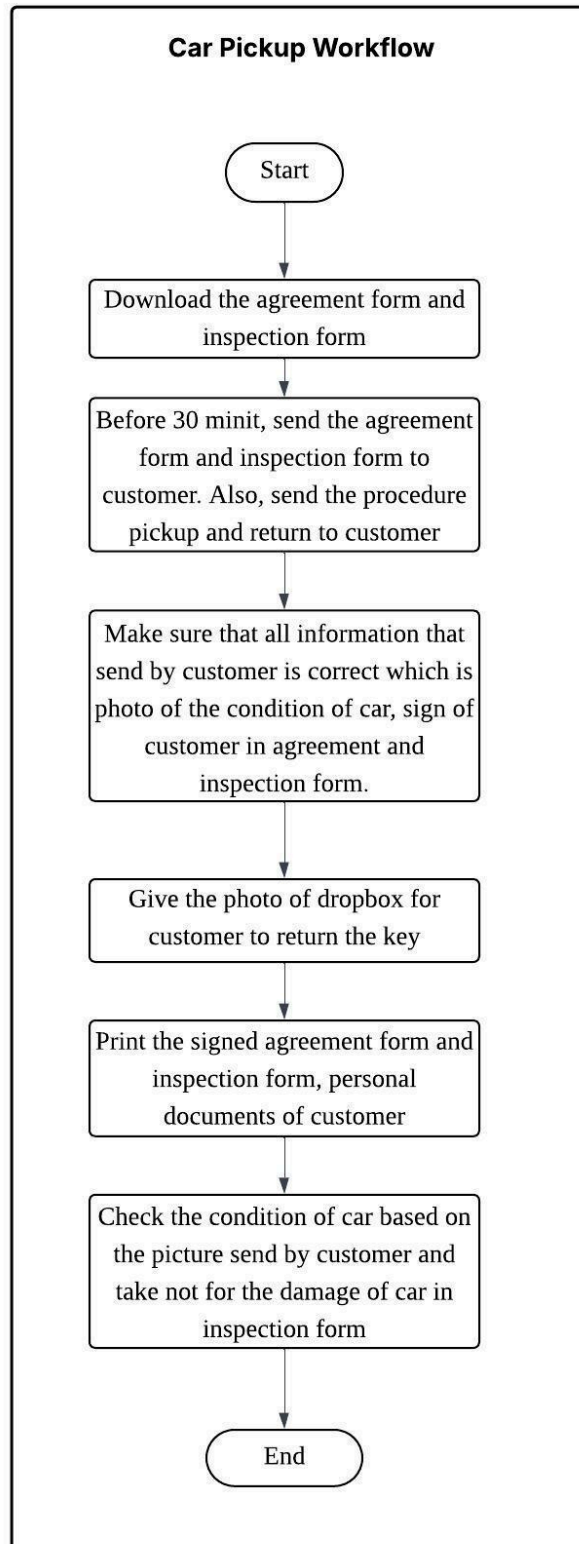


Figure 3.1.2.1 Car Pickup Workflow

3.1.3 Car Return Workflow

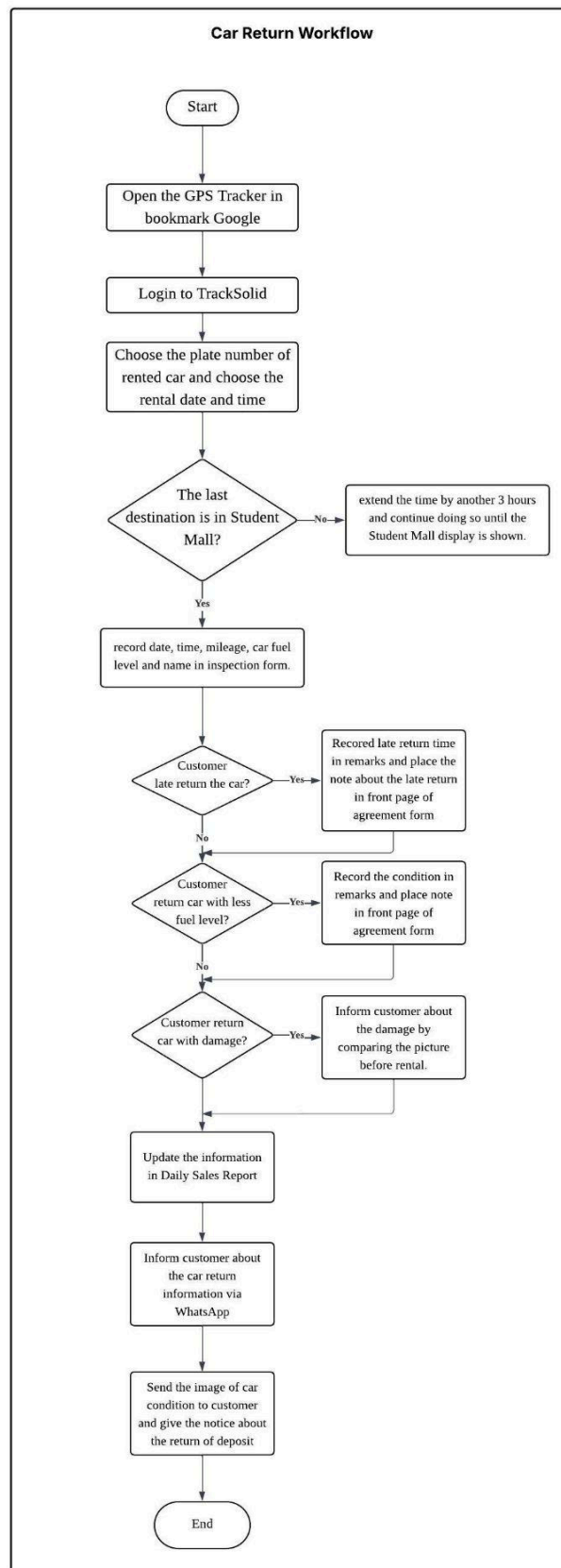


Figure 3.1.3.1 Car Return Workflow

4.0 Transaction Requirement for Current System

All data and transaction requirements are derived directly from the operational procedures documented in SOP and reflect the actual workflow used in daily vehicle rental operations. The current system maintains structured data to support daily vehicle rental operations at Hasta Travel & Tours. It stores customer information, staff involvement records, vehicle fleet details, booking transactions, payment records, loyalty information and operational reports. These data are actively used throughout the rental lifecycle. Each dataset supports operational accuracy and traceability through ensuring that all actions taken during the rental process are properly recorded and retrievable when required.

In this current system, the staff users have broad access to operational data across all modules. This unrestricted access allows staff to perform end-to-end rental operations such as eligibility verification, booking confirmation, vehicle inspection, payment tracking and reporting since there are no limitations within system-level access. In short, the system relies on operational procedures and staff responsibility to ensure proper data usage.

4.1 User Module

The current system is required to store customer and staff information to support the vehicle rental process.

4.1.1 Business Rules

1. Only customers who are UTM students or UTM staff are allowed to proceed with the vehicle rental booking process. Customers who are not UTM students or staff must be informed that the rental service is restricted and the booking cannot continue
2. Customers must submit three valid personal documents which are identification card, driving license and matric card before rental process can be completed
3. A booking cannot be confirmed if any required customer document is missing, unclear or invalid

4.1.2 Data Requirements

Customer

Customer data includes the customer's name, identification card (IC) number or passport number, phone number for Whatsapp communication, matric no, race, college, faculty, email and customer category such as UTM student or staff. In addition, the system stores three personal documents provided by customers, which are identification card, driving license and matric card. These documents are used for verification before the rental process.

Staff

Staff-related data includes the staff member name who is the person in charge (PIC) that handles the booking, key handover, inspection and return process. This information is recorded to ensure proper tracking of responsibilities and accountability.

4.1.3 Transaction Requirements

Data Entry

1. Enter customer details from booking form
2. Upload customer personal documents which are IC, driving license and matric card
3. Record staff name that handles the customer

Data Update/Deletion

1. Update customer information if correction is needed
2. Update staff name if handled by a different staff
3. Remove or replace incorrect customer documents

Data Query

1. Check customer eligibility whether the customer is UTM staff or student
2. View customer booking details
3. View customer personal documents

4.2 Fleet Module

The system maintains vehicle-related data to manage vehicle availability and condition throughout the rental process.

4.2.1 Business Rules

1. A vehicle can only be rented if its availability status is marked as available
2. A vehicle marked as not available or under restriction cannot be selected for booking

4.2.2 Data Requirements

The data includes vehicle model, vehicle plate number, colour, production year, current mileage, condition and current availability status of the vehicle. Moreover, the system also records legal and ownership-related information including date registered with Jabatan Pengangkutan Jalan Malaysia (JPJ), manufacturer, operating state and owner details such as name, identification card number, age, gender, race, citizenship, ownership, grant image, road tax image, cover note image and their respective expiry dates.

4.2.3 Transaction Requirements

Data Entry

1. Register a new vehicle with complete details
2. Record vehicle status as unavailable which means cannot be rented

Data Update/Deletion

1. Update mileage after return
2. Update availability status after booking or return

Data Queries

1. Check vehicle availability for a selected date and time
2. View latest mileage records

4.3 Booking Module

Booking-related data is required to manage reservations from confirmation until completion.

4.3.1 Business Rules

1. A booking can only be created after the customer has confirmed the rental details and selected an available vehicle
2. A booking is considered confirmed only after deposit payment is received and recorded
3. Pick up and return locations must be student malls and follow time constraints
4. Inspection forms and agreement forms must be generated for every confirmed booking
5. Vehicle key can only be handed over after required documents and payment conditions are fulfilled
6. During vehicle return, inspection data such as mileage, fuel level and condition must be recorded
7. Customer reviews related to punctuality, cleanliness, communication and overall performance must be recorded
8. Late return is calculated only after a 30-minute grace period from the scheduled return time
9. Fuel shortage penalties are applied based on the number of fuel bars missing as recorded during inspection
10. Any extension of rental duration must be recorded and charged
11. Each booking must have a staff member (PIC) assigned to handle booking, key handover, inspection or return

4.3.2 Data Requirements

Booking

Booking data includes customer name, selected vehicle model, pick up and return dates and times, pick up and return locations, rental duration and assigned staff member (PIC).

Pick Up and Return

The system stores inspection and return data also. Stored data includes inspection date and time, mileage and fuel level during pick up and return, vehicle condition notes, inspection remarks and photos from all four sides which are front, behind, left and right side, damage image if any damage exists on the vehicle Besides, information related to late returns, fuel shortages and other return-related issues is also recorded to support accurate reporting and payment deductions. The system stores customer review data for cleanliness, punctuality, communication and overall performance after a booking is completed.

4.3.3 Transaction Requirements

Data Entry

1. Enter booking details
2. Generate inspection form and agreement form

3. Record pickup inspection data
4. Record return inspection data
5. Record late return duration
6. Record fuel shortage
7. Record customer reviews

Data Update/Deletion

1. Update booking details if customer makes changes
2. Update inspection records after vehicle return
3. Update remarks for late return or fuel shortage

Data Query

1. View booking details by date
2. View inspection forms for pickup and return
3. Check booking status
4. Check return time and inspection outcome
5. View customer reviews

4.4 Payment Module

In order to ensure accurate financial tracking for each booking, payment-related data is stored also.

4.4.1 Business Rules

1. A fixed deposit must be paid before a booking can be confirmed
2. Deposit and rental payments must be recorded with the correct payment date, time and method
3. Payment status must be marked as unpaid or paid based on actual payment received
4. Penalties for late return, fuel shortage or rental extension must be deducted from deposit if applicable
5. Deposit refund is only processed after vehicle return inspection is completed
6. All payment transactions must have a corresponding receipt stored

4.4.2 Data Requirements

The payment-related data includes deposit amount, deposit payment date and time, rental payment amount, payment method such as cash or bank transfer and refund status. The system also stores the payment status whether paid or unpaid and the corresponding payment receipts for both deposit and full rental payments. In addition, data related to deductions such as late return charges, fuel shortages and extended rental charges are recorded to support deposit settlement after the vehicle is returned.

4.4.3 Transaction Requirements

Data Entry

1. Record deposit payment
2. Upload deposit receipt

3. Record full rental payment
4. Upload payment receipt
5. Record additional charges which are late return, fuel and extension charges

Data Update/Deletion

1. Update payment status from unpaid to paid
2. Update payment details if correction is needed
3. Update deduction amount after return inspection

Data Query

1. View payment history for a booking
2. View deposit and rental payment records
3. Check deduction details
4. Check refund status

4.5 Loyalty Module

The system is required to store loyalty-related data for customers.

4.5.1 Business Rules

1. A loyalty card is issued only when a customer completes a 12-hour rental
2. Subsequent loyalty stamps are awarded only for rentals with a minimum duration of 9 hours
3. Loyalty stamps must be recorded immediately after an eligible rental is completed
4. Discounts can only be claimed once the required number of stamps is achieved
5. A loyalty reward cannot be claimed more than once for the same rental

4.5.2 Data Requirements

The stored data includes loyalty card serial number, customer name and stamp count accumulated from eligible rentals. The system also records claim status when a discount is redeemed. These data ensure the loyalty rewards are rewarded and tracked correctly.

4.5.3 Transaction Requirements

Data Entry

1. Record issuance of loyalty card
2. Record first stamp for 12-hour rental
3. Record subsequent stamps for 9-hour rental

Data Update/Deletion

1. Update stamp count after each eligible rental
2. Update claim status when discount is used

Data Query

1. Check customer loyalty status
2. View stamp history

3. Check reward eligibility

4.6 Reporting Module

The system must store reporting data to support daily operational monitoring and record keeping.

4.6.1 Business Rules

1. All confirmed bookings must be recorded in the Daily Sales Report based on the rental date
2. Daily Sales Report entries must include customer details, rental details, payment details and staff involvement
3. Inspection outcomes and penalty charges must be updated in the Daily Sales Report after vehicle return
4. Daily report must accurately reflect the income while monthly report must reflect the number of sales and income

4.6.2 Data Requirements

The reporting data includes Daily Sales Report data such as report date, customer details including name, bank account number, bank name, bank account owner's name, rental details namely destination, vehicle plate number, payment method, payment amount, payment date, pick up date and time, return date and time, deposit, refund amount and staff name responsible for data entry, key handover and vehicle return. The system also records penalty-related data including return charges, fuel charges and extension charges. There is also a booking overview of all vehicles together with the vehicle details such as vehicle name, plate number and availability status. The system also has daily and monthly income reports that record the number of sales and income.

4.6.3 Transaction Requirements

Data Entry

1. Enter customer details, rental details, deposit details and staff name responsible
2. Enter penalty and deduction values such as late return charges, fuel charges and extension charges

Data Update/Deletion

1. Update the Daily Sales Report after the vehicle is returned and inspection is completed
2. Update penalty, deduction or refund values based on return inspection results
3. Update staff names if different staff members handle data entry, key handover or vehicle return

Data Query

1. View the Daily Sales Report based on a selected date
2. View daily income report showing total income for each day
3. View monthly income report showing total number of sales and income

4. View booking overview for all vehicles, including vehicle details and availability status
5. Check staff involvement for each transaction, including data entry, key handover and vehicle return

4.7 General Requirements

The current database system supports daily vehicle rental operations by storing complete and accurate data for customers, vehicles, bookings and payments. The system is able to retrieve customer records, vehicle availability, booking details and payment information quickly to help staff to carry out daily operational tasks. Manual checks such as verifying customer documents, confirming bookings and validating payments ensure the data accuracy. While for security, staff members are able to view and manage all data stored in the system. Data protection relies mainly on staff responsibility and manual supervision.

Although this shared access makes daily operations easier to manage, it also means that the system depends heavily on staff discipline and SOP compliance to prevent mistakes or misuse of data. This limitation shows the need for better role-based access control in the proposed system.

5.0 Problem Statement

The current vehicle rental management system relies on manual fill in data which creates an opportunity for entering the incorrect data or data not being consistent across the system. Data accuracy and reliability are difficult to maintain due to booking, payment, and customer records are all being managed manually across multiple platforms and each application has its own way of inputting information and result in lack of consistency.

In addition, time-wasting during the process of verification when it comes to payments, as there is a lack of real-time system integration regarding the booking, payment, fleet management system, and accounting system. Consequently, unsynchronized updates in the booking update are high probably to cause a poor user experience during using this system.

Furthermore, the absence of a centralized platform for customers to manage their own bookings, payments, and rental history increases the risk of losses and breaches of customer data. Additionally, as the usage of the current system increases, the need for users to rely upon staff or workers to operate and manage this system will become greater. At the same time, users will not have control over their own bookings or payments, which will create difficulties for them when they need to manage their rental histories or the records of their past rentals for analysis of their personal expenses.

Moreover, the current system lacks a structured and centralized mechanism to track deposits, payments, refunds, and penalties. Manually completing the financial reconciliation process will require using a spreadsheet application, which will consume a lot of time and lead to headaches from making mistakes as well. Additionally, the process that exists is not capable of meeting LHDN e-Invoice requirements, therefore could create issues with respect to meeting regulations.

Lastly, the system is lacking is the automated blacklist of customers with outstanding fines, accidents, and payment issues. The addition of this feature will improve the ability of the company to manage risks and reduce the risk of both financial and reputational loss.

6.0 Proposed Solutions

The proposed Hasta Travel Vehicle Rental Management System aims to digitalize and streamline the vehicle rental process for UTM students and staff. The system will ensure efficient user verification, secure payment handling, transparent booking management and a reward mechanism in order to encourage customer loyalty. This truly enhances the convenience, trust and also operational efficiency.

6.1 Technical Feasibility

The technical feasibility of the system proposed is reasonable because it employs web technologies that are commonly used and well-supported in application development uniformly across the web. The proposed system's development will take place using Laravel framework with PHP as the main back-end programming language, MySQL as the Database Management System (DBMS) and HTML and CSS for front-end structure and styling. Moreover, it's going to be developed and tested by using Laragon providing the same efficient development setup that will allow for an easily created and maintained stable environment. In addition, GitHub will be used for version control and collaborative development. As a result, a proper management of source codes is essential throughout the development lifecycle. Furthermore, the system design avoids complex third-party integrations, reducing technical risks while still meeting all functional requirements within the defined project scope.

6.2 Operational Feasibility

The operationally feasible system aligns well with Hasta Travel's current workflow. The proposed system called Hasta Travel Vehicle Rental Management System provides three different interfaces for different roles including customer, admin and manager. Customers can independently complete the entire rental process, including viewing the availability of vehicles, upload documents, making bookings, submitting payments, managing rental history and redeeming vouchers through this centralized system. As a result, this will reduce relying on manual staff communication and improve overall service efficiency. The admin can handle customer verification, fleet details, booking approval requests, payments, blacklist data, and loyalty points from their admin website. Manager roles have similar functions with admin but have an additional function which is reporting. Managers can view financial reports, earning summaries and booking statistics. Moreover, owing to a friendly interface with a web-based system already familiar to their staff, there is no need for extensive training to implement this system in their day-to-day operations.

6.3 Economic Feasibility

A Cost-Benefit Analysis (CBA) has been carried out to evaluate the economic feasibility of the Hasta Travel Vehicle Rental Management System.

Table 4.3.1: Estimated Cost

Estimated Cost	
Hardware	RM5,000
Software	RM5,300
Training	RM1,400
Consultant	RM1,600 per year
Maintenance	RM3,000 per year

Table 4.3.2: Estimated Benefits

Estimated Benefits	
Reduced Errors	RM4,100
Increased Online Booking	RM3,500
Staff Time Saving	RM2,800
Improved Customer Retention & Satisfaction	RM4,200

Table 4.3.3: Assumptions

Assumptions	
Discount rate	10%
Sensitivity factor (cost)	1.1
Sensitivity factor (benefits)	1.1
Annual increase in benefits	5%
Annual increase in production cost	5%

Table 4.3.3: Cost-Benefit Analysis (CBA)

Costs	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Development Costs						
Hardware	5,500					
Software	5,830					
Training	1,540					
Total	12,870					
Production Costs						

Consultant		1,760	1,848	1,940	2,037	2,139
Maintenance		3,300	3,465	3,638	3,820	4,011
Annual Production Costs		5,060	5,313	5,579	5,858	6,150
(Present Value)		4,600	4,391	4,191	4,001	3,819
Accumulated Costs		17,470	21,861	26,052	30,053	33,872
Benefits						
Reduced Errors		4,510	4,736	4,972	5,221	5,482
Increased Online Booking		3,850	4,043	4,245	4,457	4,680
Staff Time Saving		3,080	3,234	3,396	3,565	3,744
Improved Customer Retention & Satisfaction		4,620	4,851	5,094	5,348	5,616
Annual Benefits		16,060	16,863	17,706	18,591	19,251
(Present Value)		14,600	13,936	13,303	12,698	12,121
Accumulated benefits (Present Value)		14,600	28,536	41,839	54,537	66,658
Gain or Loss		-2,870	6,675	15,787	24,484	32,787
Probability Index	2.55					

7.0 Proposed System

7.1 Mission Statement

The proposed Hasta Travel Vehicle Rental Management System aims to digitalize and streamline the vehicle rental process for UTM students and staff. The system will ensure efficient user verification, secure payment handling, transparent booking management and a reward mechanism in order to encourage customer loyalty. This truly enhances the convenience, trust and also operational efficiency.

7.2 Overview

The to-be system is a web-based vehicle rental management platform designed for Universiti Teknologi Malaysia (UTM) users. It will handle every stage of the vehicle rental process, which is from user registration and verification to vehicle booking, payment and loyalty rewards. The system will also allow the administrators to manage user data, vehicle information, bookings, payments and blacklisted accounts. Other than that, customers can easily make bookings and view vehicle availability in real-time via this system.

The main objectives of this proposed system are to simplify the vehicle rental and verification process for UTM students and staff. Moreover, it will ensure only verified and eligible users which are non-blacklisted can access the system. Most importantly, this system plans to provide users real-time vehicle availability using the booking calendar so the users need not to ask the staff through Whatsapp about the availability of the vehicles. With vouchers rewarded to loyal customers, this system aims to encourage repeated use from customers. Lastly, it truly helps staff to efficiently verify documents, approve bookings, manage blacklists and monitor transactions.

7.3 Objectives

The objectives of this project have been clearly defined to ensure that the proposed database system effectively solves the problems faced in the current vehicle rental booking process and meet the requirements of clients.

The objectives of the proposed system are as follows:

1. To create a digital platform that allows customers to review vehicle details, prices, and available time, and complete the entire booking process without manual booking through WhatsApp.
2. To reduce manual work by enabling staff to create, edit, delete, and retrieve customer, vehicle, booking, and payment data efficiently.
3. To improve the accuracy of data such as booking details, payment records, and vehicle availability by applying the use of a centralized database.
4. To help admin manage rental operations more effectively, including booking confirmation, document verification, payment tracking, and vehicle status updates.
5. To provide automatic generated reports that allow management to view and analyse rental activities and track financial flows clearly in purpose to support decision-making.

7.4 Scope

This section describes the scope of the proposed vehicle rental booking system. This defines the features included in the vehicle rental booking system and the features that are not included in this project.

7.4.1 In Scope

7.4.1.1 User Management

While registering the account, users need to fill in their email and have to create a username and password which are used for login purposes. Users can login into the system and have a look at the vehicle models and respective details. However, before proceeding to book a vehicle, it is mandatory for the users to fill in all the information required in the profile section. For example, users must enter their name, matric number, faculty, address and other personal data. Also, users need to upload their identity card or passport, matric card and also driving license for verification. Besides, emergency contact details are required. Manual verification will be done by staff before they can book the vehicle.

If a user is blacklisted because of outstanding payments, they will still be able to log in and view the vehicle models and details of the vehicle. However, they would be restricted from accessing other system features. Once the blacklisted user wants to rent a vehicle, they will be redirected to a payment page to clear their outstanding payments.

7.4.1.2 Vehicle Management

The system will display vehicle models handled under the company and their respective details such as price, colour, engine, power and number of seats. Furthermore, there will be real-time availability shown via an interactive calendar. The calendar highlights when the vehicle is booked and enforces a 1-hour cooling period between bookings.

7.4.1.3 Booking Management

The system will allow customers to view the availability of vehicles and make bookings easily. When filling the booking form, the customer can directly choose their rental time on the calendar which could reduce the redundancy as the customer needs not to refer back to another page for the availability. Other than that, if a customer chooses other locations instead of Student Mall for pick up location and return location, extra fee would be charged. The booking fee is automatically calculated based on rental period, pick up point, deposit of RM50 per booking and voucher used. After the staff has verified the payment, customers will receive a confirmation notification through their email and also on the website.

Before picking up the vehicle, customers would have to take photos of the vehicle before rental, fill in the current fuel level and upload in the system besides the agreement form. Then, the system would automatically start the countdown after customers finish uploading the documents. On the other hand, for returning, the customers are required to upload the photos of the vehicle and the inspection form. They can even give feedback and ratings on this rental whether about the service or the vehicle.

On the other hand, if the customer wishes to cancel the booking, they have to make it 24 hours before picking up the vehicle. Then, they would get the refund including the deposit. Nonetheless, if the customer only manages to cancel the booking within 24 hours before the pick up, they would not get their refund back.

7.4.1.4 Payment and Penalty Handling

The system supports a payment method which is QR payment. Payment receipts must be uploaded for verification. The system will also automatically calculate and apply penalties for late return. For this reason, there is a feature that can track and start the time countdown once the customer gets the vehicle key. Users have a 30-minute grace period after return time, after this period, regardless of whether the vehicle is a car or a motorcycle, an additional fine of RM50 will be charged for each hour of overdue return.

7.4.1.5 Loyalty Rewards

The user will get a loyalty stamp when they make a booking that is more than 9 hours. When the user gets 3 stamps, the user would be allowed to redeem discount vouchers such as 12-hour free booking that can be used on a random day. Vouchers are viewable and redeemable through the user's account.

7.4.1.6 Admin Dashboard

The admin dashboard would display the important activity and information needed to be reviewed by the admin such as total revenue, active rental, pending task, quick links and action center. While the manager dashboard contains more visualized graphs that show performance trends, top rented models, customer analytics, fleet and booking status. On the other hand, both manager and admin can verify user documents, approve or reject the users bookings besides updating vehicle availability by having the overview of all vehicles. Moreover, they can manage the blacklists and track the payments.

7.4.2 Out of Scope

7.4.2.1 Vehicle Insurance Processing

The system does not handle vehicle insurance claims or policy renewals. All insurance-related matters would remain under Hasta Travels external insurance provider to avoid integration with third-party insurance APIs and confidential legal data.

7.4.2.2 Integration with External Banking APIs

Although the system accepts payment receipts, it does not connect directly to bank systems. Nevertheless, all payment verifications are done manually by staff to avoid the complexity of financial API compliance.

7.4.2.3 Automated Staff Assignment

If a customer requests delivery, manual staff assignment will be used. The system will not automatically schedule or dispatch drivers based on the route optimization.

7.4.2.4 Real-time vehicle tracking using GPS

The system does not include real-time vehicle tracking using GPS. Vehicle location monitoring and live tracking are not implemented to avoid additional hardware requirements and system complexity.

7.5 System Boundaries

The system serves as the main platform which connects customers which are UTM students or staff with Hasta Travel & Tours admin and manager. It interacts with external entities only through limited manual data exchange.

Table 7.5.1 Interaction between entities and system

Entities	Description
Customer (UTM student or staff)	Registers, logins, update personal details, uploads verification documents, views vehicle models, prices and details, checks availability, books vehicles, makes payments, fills pickup and return form, earns loyalty stamp, redeem vouchers, cancels or modifies bookings
Admin	Verifies users documents, approves or rejects bookings, updates vehicle availability and status, handles payment verification, adds or removes users from blacklist, manages vouchers
Manager	Views faculty and residential college booking statistics, reviews financial reports, manages admin

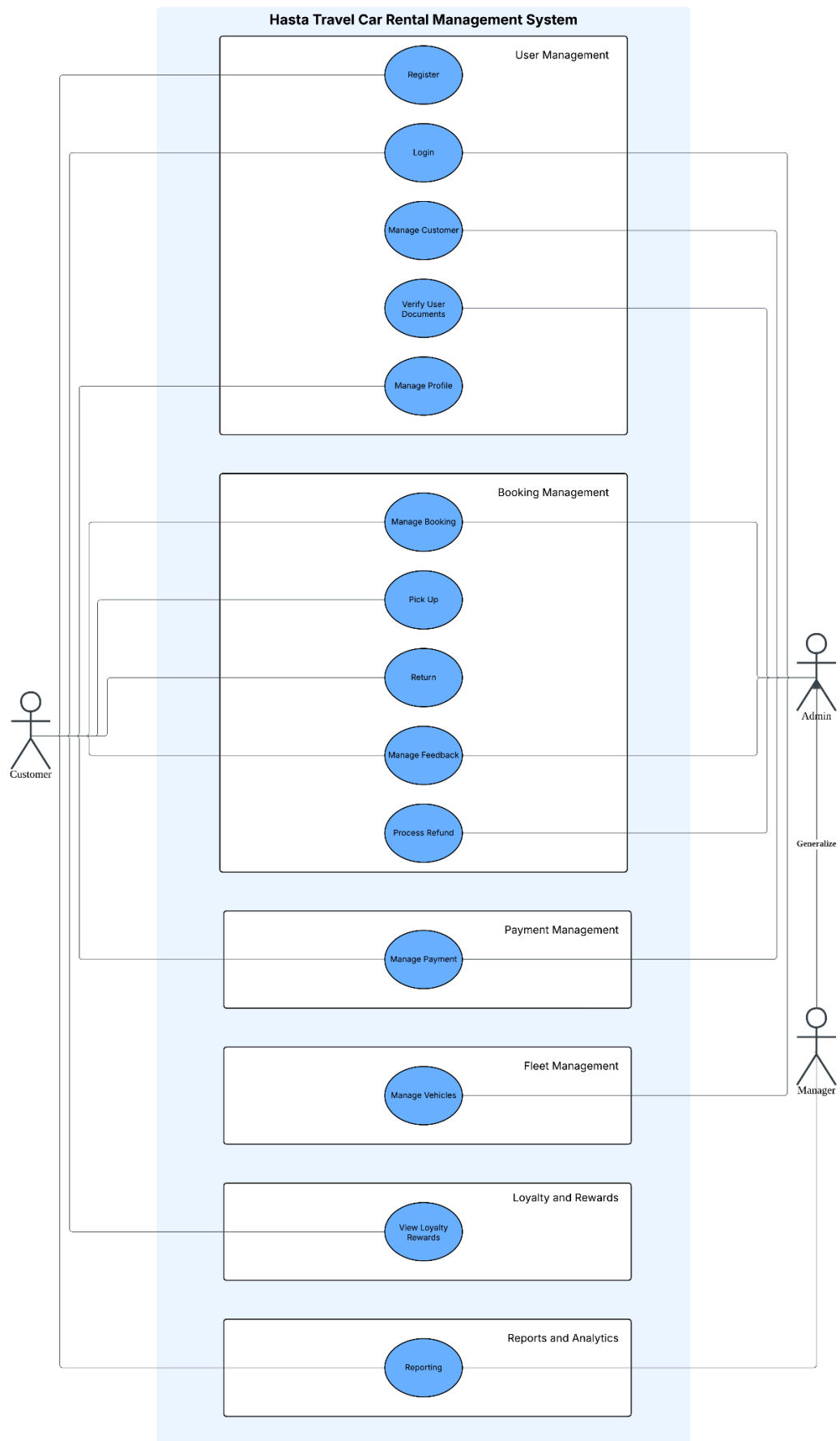


Figure 7.5.1 Use Case Diagram

7.6 Proposed Technologies

Table 7.6.1 Technology used for Proposed System

Component	Proposed Technology
Frontend	HTML, CSS and Javascript
Backend	PHP
Database	MySQL
Hosting	Laragon
Security	User authentication, blacklist access control and input validation

8.0 Transaction Requirement for Proposed System

In the proposed system, user views are controlled through role-based access. Each user is assigned a role, namely Customer, Admin or Manager. The system restricts data visibility and system functions according to the assigned role so the users only access information relevant to their responsibilities.

Customers have a personalized view that allows them to manage their own profile, upload personal documents, create and track bookings, view payment and deposit records, monitor loyalty stamps, redeem vouchers and submit feedback after rental completion. Customers are restricted from accessing other users' data or administrative functions.

Admins are provided with access to operational data required for daily rental management. This includes customer records for verification, fleet availability, booking transactions, payment verification, pickup and return inspections. Admins can manage day-to-day activities but have limited authority over system configuration and role management.

Lastly, for managers, they have full system visibility and control. They can access all modules, manage user roles, create admin accounts, monitor financial performance through reports and dashboards. This role-based user view structure improves data security, accountability and operational efficiency compared to the current system.

8.1 User Management Module

User Module serves as the centralized identity management component of the Hasta Travel & Tours Rental Management System. It securely manages all registered entities including Customers, Admin and Manager. Moreover, it controls authentication, authorization and profile information.

8.1.1 Business Rules

1. All new public users are registered with the customer role by default
2. Customers must complete their specific profile data which are driving license, Identification Card (IC) or passport and matric card before they are permitted to book a vehicle
3. Uploaded documents require manual approval by an admin or manager
4. Admins or managers can approve or reject these documents and a reason is needed when rejection occurs
5. When the customers change the IC number or matric no, they have to re-upload the related documents
6. Booking privileges are blocked until documents are approved
7. Customers with outstanding penalties are marked as blacklisted and cannot make new bookings until they upload a payment receipt which is then verified by an admin

8.1.2 Data Requirements

Users

Hasta Travel Vehicle Rental Management System maintains user accounts for all individuals accessing the platform. Each user is uniquely identified by ID and must register with a valid

and unique email address with password. Stored data includes, ID, email, username, password, role and account status.

The user role attribute determines the level of system access and dashboard features available after login. Roles include customer, admin and manager. User roles are not fixed at registration and may be updated by authorized personnel. A Manager has the authority to update a user's role directly within the User table, for instance, upgrading a Customer to a Manager. This design enables flexible role management without requiring the creation of new user accounts.

Customers

Customer records represent Universiti Teknologi Malaysia (UTM) students or staff using the booking service. Each customer is uniquely identified by a customer ID. Stored data includes customer ID, full name, nationality, matric or staff card image, matric or staff number, identity card or passport image, identity card or passport number, gender, faculty, residential college, address, phone number, driving license image, driving license expiry date, bank name, bank account, emergency contact information including name, relationship with customer and phone number, document verification status and customer status. Customer data is linked to the corresponding user account.

Admin

Admin members are system users with additional privileges and created through Manager actions. Each admin is uniquely identified by an admin ID. Stored data includes admin ID, phone number, position and admin status. Admin accounts are linked to user credentials for authentication.

8.1.3 Transaction Requirements

Data Entry

1. Register a new user account with default role Customer
2. Promote an existing user to Manager by updating the role attribute
3. Create Admin account through Manager authorization
4. Enter user credentials and profile information

Data Update/Deletion

1. Update a user's role, for example, Customer to Manager
2. Activate or deactivate Admin privileges
3. Update user credentials and profile information
4. Update document verification status from pending to approved or rejected
5. Deactivate user accounts without deleting historical records

Data Queries

1. Retrieve user role during login to determine access rights
2. Load role-specific dashboard and permissions
3. Check email uniqueness during user registration

4. Handle password reset requests using registered email
5. Verify Manager authorization before allowing admin creation
6. List users based on role for manager management

8.2 Fleet Management Module

This section outlines the vehicle module transaction requirements, which manage vehicle information, availability and status in the Hasta Travel & Tours Vehicle Rental Management System.

8.2.1 Business Rules

1. The system designates vehicles in two primary states which are 'active' for those currently in the operational fleet and 'retired' for those permanently inactive or the owner does not want to rent the vehicle anymore
2. Vehicle availability is tracked dynamically which 'available' indicates readiness for booking while 'unavailable' indicates the vehicle is currently rented or disabled
3. A maintenance status overrides standard availability which prevents bookings when a vehicle is undergoing repairs
4. Rental prices are defined individually for each vehicle across standard durations which are one hour, three hours, five hours, twelve hours and one day. The system also applies different pricing tiers for weekends versus weekdays to accommodate demand fluctuations
5. The system records detailed owner information for each vehicle
6. Critical documents such as the vehicle grant, road tax and insurance cover note are uploaded and tracked digitally
7. The system actively tracks expiry dates for road tax, insurance and other recurring obligations. Admin can set due date and receive alerts for upcoming renewals
8. Service records log details of maintenance work, vendor name, cost and date while meter updates records tracks the odometer readings periodically to monitor vehicle usage

8.2.2 Data Requirements

Vehicle

Each vehicle is uniquely identified by vehicle plate number. Stored data includes general information namely, plate number, model, colour, seat, vehicle mileage, production year, engine, engine CC, fuel type, transmission type, drive type, maximum speed, torque and power, pricing rules, manufacturer, operating state and owner details such as name, identity card number, age, gender, race, citizenship, grant image path, roadtax image path, cover note image path and their respective expiry dates.

Operational data includes status, availability status and maintenance status. Status defines whether the vehicle is a valid member of the fleet that can be rented or not while availability status and maintenance status are used to control booking and fleet management activities.

Vehicle Photos

The system stores multiple photos for each vehicle to provide visual reference for system users. Each photo record is uniquely identified by a photo ID and is linked to a specific vehicle plate number. Stored data includes photo ID, vehicle plate number and label. This separation allows multiple images to be associated with a single vehicle without duplicating vehicle data.

Renewal Reminder

The renewal reminder records are used to manage expiry-related vehicle documents such as road tax and insurance. Each renewal reminder is uniquely identified by a reminder ID and is linked to a specific vehicle. The data stored for each renewal reminder includes reminder ID, service type (such as insurance or road tax), due date, reminder notes and reminder status. This separation allows a vehicle to have multiple renewal reminders and enables the system to track upcoming expiries without affecting core vehicle data.

Service History

The system stores maintenance and meter update information for vehicles in the system. Each service history record is uniquely identified by a service ID and is linked to a specific vehicle. Stored data includes service ID, vehicle name, completed date, current odometer reading, previous odometer reading, previous odometer update date, service type, service vendor, service description and service amount. This supports multiple service and meter update records for the same vehicle, allowing the system to maintain a complete history of vehicle maintenance and mileage updates.

8.2.3 Transaction Requirements

Data Entry

1. Register a new vehicle in the admin or manager site of the system with complete technical, legal and operational details
2. Upload one or more photos for a vehicle
3. Define hourly rental prices for a vehicle based on different time durations
4. Insert a renewal reminder record and record the service type, due date and optional notes
5. Insert a new service history record when a vehicle undergoes maintenance
6. Insert a new meter history record when the vehicle odometer is updated such as after rental return or inspection

Data Update/Deletion

1. Update vehicle details such as availability status, mileage or maintenance status
2. Update or replace vehicle photos
3. Modify the pricing rate
4. Deactivate a vehicle from the system without removing historical booking data
5. Update the renewal reminder status, due date and note when the expiry date is approaching or when a reminder has been processed

6. Delete a renewal reminder record if it is entered incorrectly or no longer required
7. Update service history details if corrections are required
8. Update previous odometer information automatically when a new meter update record is added
9. Delete a service history record if it was entered incorrectly without affecting other service records

Data Queries

1. Retrieve a list of available vehicles based on date and time
2. Retrieve vehicle details and associated photos for display
3. Retrieve applicable rental duration pricing for a selected vehicle
4. Check vehicle availability before confirming a booking
5. List vehicles by status (available, unavailable, under maintenance)
6. Retrieve vehicle usage data for reporting and analysis
7. Retrieve renewal reminders that are approaching the due date
8. Retrieve renewal reminders based by service type and based on vehicle plate and reminder status
9. Retrieve the complete service history of a vehicle based on vehicle plate number
10. Retrieve meter update records to track odometer changes over time

8.3 Booking Management Module

The Booking Module manages the complete vehicle rental lifecycle, from reservation creation to vehicle pick up and return. It coordinates booking details, rental periods, vehicle assignment and booking status updates. This module ensures that vehicles are allocated correctly, rental rules are enforced and booking records are maintained for operational and reporting purposes. It allows customers to create, manage and track vehicle bookings while allowing admin and manager to verify, approve and monitor booking activities.

8.3.1 Business Rules

1. Customers must have a complete profile with all approved uploaded documents before they can initiate a booking
2. The system enforces a conflict check to strictly prevent any booking that might overlap with an existing pickup or return for the selected vehicle
3. The system applies additional fees for pickups or returns that occur at designated surcharge locations such as outside campus or during specific surcharge times. Surcharge rules are configurable via the admin and manager panel
4. Customers are only permitted to cancel a confirmed booking if the scheduled pickup time is more than 24 hours away.
5. Cancellations are eligible for a refund which is processed manually within 3 to 5 business days
6. If a booking using a voucher is cancelled, the voucher will be automatically restored for future use

7. Both pickup and return processes require a digital handover where the customer must upload four angle photos of the vehicle, a dashboard photo and note any damage
8. Fuel levels are recorded at both ends of the transaction to calculate penalties
9. Customers must digitally acknowledge the rental agreement at pickup
10. Customers are required to fill the ratings and feedback at return
11. Late return is calculated only after a 30-minute grace period from the scheduled return time and RM50 would be charged per hour
12. Fuel shortage penalties are applied based on the number of fuel bars missing as recorded during inspection which are RM10 per missing fuel bar
13. The penalties are subtracted from the deposit. If the penalty is less than deposit, the remaining deposit is refundable

8.3.2 Data Requirements

Booking

Each rental is uniquely identified by a rental ID. Stored booking data includes rental ID, booking status, withdrawal receipt, withdrawal status, destination, invoice, payment status, fuel penalty and late penalty. Booking status indicates the current stage of the rental process such as pending, verified, renting, done and rejected. Each rental has one payment and one voucher. Each booking is associated with one vehicle and a customer can make multiple bookings.

Pick Up

Pick up records capture information related to vehicle handover to the customer. Each pick up record is uniquely identified by a pick up ID and linked to a rental. Stored data includes pick up ID, rental start date and time, pick up location, vehicle condition images at pick up, dashboard image, damage image if any damage exists on vehicle before pick up, description about damage part, fuel level at pick up, agreement form, acknowledgement status and pick up report. This information ensures proper documentation of the vehicle condition before the rental begins.

Return

Return records store information related to vehicle return at the end of rental period. Each return record is uniquely identified by a return ID and linked to a rental. Stored data includes return ID, rental end date and time, return location, vehicle condition images at return, dashboard image, damage image if any damage exists in the rental period, damage remarks, fuel level at return, service rating, vehicle condition rating, feedback, acknowledgement of the images for different views and return report.

8.3.3 Transaction Requirements

Data Entry

1. Create a new booking by selecting a vehicle, rental period and destination
2. Record vehicle pickup details during handover
3. Record vehicle return details upon rental completion

Data Update/Deletion

1. Update booking status throughout the rental lifecycle
2. Update late penalty if the customer returns the vehicle late
3. Update fuel penalty when the return fuel level is less than pickup fuel level
4. Cancel or close bookings without deleting historical records

Data Queries

1. Retrieve booking details for a specific customer or vehicle
2. Check vehicle availability for a selected rental period
3. Retrieve pickup and return records linked to a booking
4. List verified, in progress, completed or cancelled bookings
5. Retrieve booking history for customer dashboards

8.4 Payment Management Module

This module manages all financial transactions within the Hasta Travel & Tours Vehicle Rental Management System. It handles rental payments, deposits, penalties and refunds while supporting manual payment verification by administrators or managers. It ensures that all payments are properly recorded, verified and linked to the relevant booking and customer records.

8.4.1 Business Rules

1. A deposit is mandatory and collected for every single booking iteration
2. Admin must visually verify the uploaded receipt image before a booking can be confirmed
3. Rejected receipts will trigger a re-upload request or booking cancellation. Only “incorrect amount” triggers a refund workflow while other reasons typically require re-upload
4. The system records the specific admin ID who verified each payment for accountability

8.4.2 Data Requirements

Each payment transaction is uniquely identified by a payment ID. Stored data includes payment ID, payment amount, deposit amount, penalty amount, payment type, payment method, receipt image and refund amount.

The payment type indicates whether the transaction is for a booking payment or a penalty charge. The payment method is QR payment. The verification status reflects whether the

payment is pending, approved or rejected. Each payment record is linked to the corresponding customer and verified by an authorized admin or manager.

8.4.3 Transaction Requirements

Data Entry

1. Record a new payment for a booking or penalty
2. Upload payment receipt image for verification
3. Record deposit amount associated with a booking
4. Assign verifying staff ID during payment verification

Data Update/Deletion

1. Update payment verification status whether approved or rejected
2. Record rejection reason for failed payment verification
3. Update deposit balance after successful verification

Data Queries

1. Retrieve payment details for a specific booking or customer
2. Retrieve pending payments awaiting verification
3. Retrieve verified payments processed by a specific admin or manager
4. Retrieve deposit and penalty payment records
5. Check payment verification status before confirming booking
6. Generate payment summaries for reporting and auditing purposes

8.5 Loyalty and Rewards Module

Loyalty and Rewards Module manages customer incentive programs within the Hasta Travel & Tours Vehicle Rental Management System. It tracks customer rental activity, awards loyalty stamps and manages reward vouchers. This module aims to encourage repeat usage by rewarding loyal customers with discounts and free rental benefits.

8.5.1 Business Rules

1. Customers earn 1 loyalty stamp for every completed rental transaction with a duration of 9 hours or more
2. Customers can redeem collected stamps for vouchers based on defined tiers. Redemption immediately deducts the stamp count from the customer's balance
3. Vouchers can optionally grant a percentage discount or specific free hours
4. A voucher can only be used once
5. If the booking is cancelled or rejected, the voucher will be automatically restored for future use
6. Admin and manager can assign voucher to the customers

8.5.2 Data Requirements

Loyalty

For the loyalty record, it is uniquely identified by a loyalty ID and is linked to a customer. Stored data includes loyalty ID, loyalty type, stamp description, rental counter, number of stamps earned.

The loyalty type indicates the nature of the reward activity such as rental-based stamps. The rental counter would track the number of qualifying rentals while the stamp count reflects accumulated loyalty progress. Loyalty records are generated automatically based on completed bookings that meet eligibility criteria or when the customers redeem the voucher.

Voucher

Each voucher is uniquely identified by a voucher ID and is linked to a customer. Stored data includes voucher ID, voucher code, voucher description, discount percentage, free rental hours, expiry date and voucher status.

Vouchers are not issued automatically. Customers may manually redeem vouchers through the system once they have sufficient loyalty stamps. Moreover, Admin and Manager can issue the vouchers as part of promotions or special cases. Voucher status indicates whether a voucher is active, used or expired.

8.5.3 Transaction Requirements

Data Entry

1. Generate loyalty records when a qualifying rental is completed or when a customer redeems the voucher
2. Create voucher records through customer redemption
3. Create vouchers manually by Admin or Manager to specific customers
4. Assign voucher codes and reward attributes

Data Update/Deletion

1. Update loyalty stamp count after eligible bookings
2. Increment rental counter after eligible bookings
3. Deduct rental counter upon successful redemption
4. Update voucher description, expiry date or status by Admin or Manager
5. Deactivate vouchers without deleting historical records

Data Queries

1. Retrieve loyalty stamp progress for a specific customer
2. Retrieve active, used and expired vouchers for a customer
3. Validate voucher eligibility during booking
4. Retrieve vouchers created or managed by Admin or Manager
5. Generate loyalty and voucher usage for reporting purposes

8.6 Report and Analytics Module

The Report and Analytics Module provides administrative and managerial insights by analyzing and summarizing data collected from existing system modules. It retrieves and processes data from the User, Vehicle, Booking and Payment modules to generate meaningful reports and statistical summaries for operational monitoring and decision-making purposes.

8.6.1 Business Rules

1. Admin dashboard focuses on operational or daily tasks while manager dashboard focuses on strategic performance
2. Revenue is aggregated into Daily (Revenue vs Expenses) and Monthly (Total Sales Count and Amount) views
3. The dashboard has fleet status overview which shows whether the fleet are available, rented or under maintenance
4. Feedback Scores view the vehicle rating especially those ratings below 3 stars to alert admin and manager

8.6.2 Data Requirements

Sales and Revenue Data

Sales and revenue information is obtained from verified payment records in the Payment Module, including payment amount, deposit value, payment type and transaction timestamps. These records allow revenue to be aggregated and analyzed according to different time periods such as daily, monthly and yearly summaries. Moreover, it can be categorized based on vehicle details and customer types retrieved from the Fleet and Use Modules.

Fleet Performance

Fleet performance data is generated by analyzing vehicle availability and booking information stored in the Fleet and Booking Modules. Maintenance performance and downtime are assessed using vehicle maintenance and availability status fields. Meanwhile for the vehicle popularity metrics, they are derived from booking frequency data to identify high-demand models within the fleet.

Customer Feedback

Customer feedback data is sourced from feedback records submitted after booking completion. This includes numerical service and vehicle ratings, along with textual feedback provided by customers. Feedback data is retrieved to support trend analysis and service quality evaluation.

8.6.3 Transaction Requirements

Data Entry

1. Generate monthly and yearly sales reports using verified payment data
2. Generate fleet utilization analysis comparing booked hours against total available hours
3. Generate customer feedback summaries after rental completion

Data Update/Deletion

1. Regenerate reports dynamically based on updated criteria
2. Update report parameters such as data range filters to regenerate views dynamically

Data Queries

1. Retrieve total revenue for a specific date range
2. Retrieve the average vehicle rating scores
3. List all feedback entries with low ratings which is less than 3 stars for manager review
4. Export report data in PDF and Excel format for offline analysis

9.0 Benefit and Summary of Proposed System

The proposed vehicle rental booking system is beneficial to customers and management.

9.1 Benefits

The system improves the user experience. The customer is enabled to view the details of the vehicle and the rent time available without waiting for a response via WhatsApp. The online system enables the customer to book and upload payment receipt documents and receive booking confirmations via email.

This system decreases the work done manually by humans because all the data from the customers, the process of booking and payment, as well as the vehicles' data, are controlled by the admin and the manager from the central platform. This decreases the risk of committing errors manually.

Additionally, the system promotes a smooth interface for communication to take place between the customers and the staff. The system sends alerts through mail and web page notifications to inform the customers of key activities such as submission and approval of bookings.

In fact, the system has also improved the accuracy and management of the data. This is because the customer, booking, payment, and vehicle details are now organized through the central database.

Moreover, the automatically produced sales reports allow an organization's management to monitor the activities and financial performance quite easily and clearly.

9.2 Summary of Proposed System

The proposed system is a vehicle booking system developed for Hasta Travel to replace their current vehicle booking process done manually. The new system allows customers to access vehicle data, make bookings, upload payment receipts, and receive booking confirmations in one interface. For the staff and management, the system will offer the ability to handle customer records, bookings, payments, as well as the status of the vehicles in a more efficient manner. This is achieved through the automation of basic processes.

10.0 Project Planning

10.1 Human Resource

To ensure the project could be completed smoothly,our team decided to make an organisation chart to distribute positions for each group member. In this organization chart,it represents the structure of our system development project team.

Hasta Travel Vehicle Rental Management System Organisation Chart

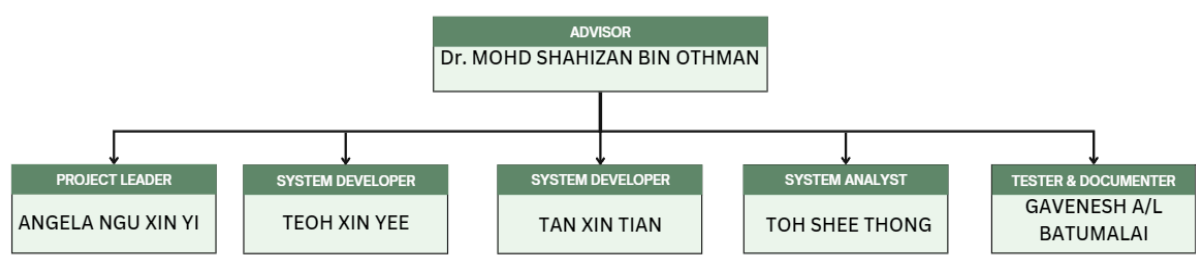


Figure 10.1.1 Organisation Chart

Table 10.1.1 Roles and responsibility table

Name	Role	Main responsibilities
Angela Ngu Xin Yi	Project Leader	- Leads and oversees the overall project progress, managing timeline and milestone tracking - Coordinates communication between the client and team to ensure the system aligns with client expectations - Provides guidance and resolves conflicts within the team
Teoh Xin Yee	System Developer (Backend)	- Implements server side logic and manages database connection and logic - Collaborate with other developers to ensure system functionality
Tan Xin Tian	System Developer (Frontend)	- Create wireframes and prototypes - Designs and develops user interface components - Ensures responsive design and smooth user experience

Toh Shee Thong	System Analyst	• Collect and gather user requirements • Prepares problem statements and use cases • Act as a middle between technical teams and the client, and validates functional requirements
Gavenesh A/L Batumalai	Tester & Documenter	• Works with developers to verify bug fixes and re-test • Ensures the final product meets quality standards and client requirements • Giving feedback and suggestions on system performance and usability

10.2 Work Breakdown Structure (WBS)

Our team making this Work Breakdown Structure (WBS) is to break down a large project into smaller and simpler parts and tasks. It could organise the total work into a hierarchical structure of tasks, which helps our teams understand the project scopes, build a project plan, and track progress effectively.

The Work Breakdown Structure (WBS) illustrates the main phases of the Vehicle Rental Management System project, including Planning, Analysis, Design, Development, Testing, and Documentation. Each phase is further divided into specific tasks to guide the project's progress.

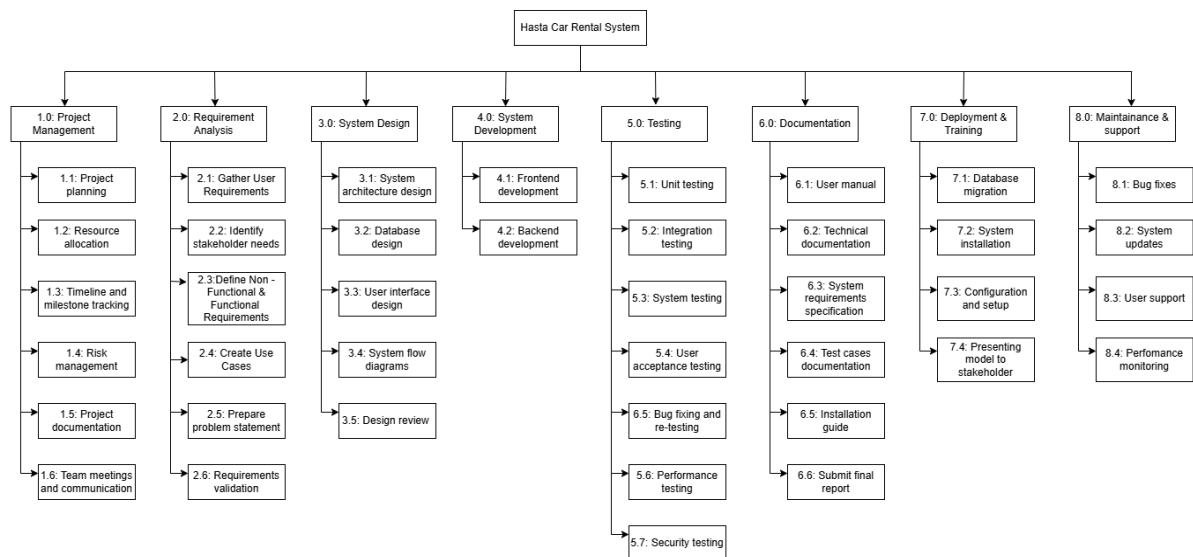


Figure 10.2.1 WBS For Hasta Travel Vehicle Rental Management System

10.3 Gantt Chart

The purpose of our team making this Gantt Chart is to provide a clear, visual overview of a project's timeline, helping teams manage, coordinate, and track progress. This Chart is suitable for planning a big project because it can divide a complex project into multiple tasks and it could help our team with good time management and keeping our project on track by following the progress shown in the Gantt Chart.

https://docs.google.com/spreadsheets/d/1w4DH-ahgGoHCECic6wYNo_erNm6wGwzVYrEwOv4dp9A/edit?usp=sharing

Figure 10.3.1 Gantt Chart For Hasta Travel Vehicle Rental Management System

11.0 Summary

The proposed system for the Hasta Travel Vehicle Rental Management System is intended to solve the problems that currently exist in the booking of vehicle rentals. The manual process of booking carried out via WhatsApp has shown to cause some delays and difficulties in handling bookings and payment data. Thus, the proposed system has been developed to solve these problems.

The proposed system enables the customer, booking, payment, and vehicle data to be stored in the same system. The proposed system helps reduce the chances of human errors occurring when handling the data manually. Customers are able to access the vehicle details, make requests for bookings, attach the required documents, and also obtain updates using the email and the web page.

The system is also beneficial to staff and management in managing vehicle rental activities. Staff are in a position to update booking information, check documents, and change vehicle information in an organized manner. The basic report functionality enables management to track vehicle rental operations and financial documents.

In conclusion, the new system will therefore serve as an effective solution for enhancing communication, data, and operations for both the customers and internal users of Hasta Travel.

12.0 Glossary

Term	Definition
Admin	An authorized system user with elevated privileges to manage day-to-day operations including customer verification, booking approval, payment verification, and fleet management.
Manager	The highest-level system user with full access to all modules, including financial reports, user role management, and system configuration.
Application Programming Interface (API)	A set of protocols and tools that allow different software applications to communicate with each other.
Blacklist	A status assigned to customers who have outstanding payments, penalties, or other violations, restricting them from making new bookings until issues are resolved.
Cost-Benefit Analysis (CBA)	A financial analysis method used to evaluate the economic feasibility of the proposed system by comparing projected costs against expected benefits.
Cubic Capacity (CC)	The engine displacement or volume of an engine's cylinders, measured in cubic centimeters.
Database Management System (DBMS)	Software that handles the storage, retrieval, and management of data in a database. MySQL is used in this project.
Driveo	The internal employee admin management system currently used by Hasta Travel to check vehicle availability and fleet information.
Grace Period	A 30-minute buffer time after the scheduled return time before late penalties are applied.
Global Positioning System (GPS)	A satellite-based navigation system used for vehicle tracking (currently via Tracksolid Pro).
Identification Card (IC)	Malaysian national identity card (MyKad) required for customer verification.

Jabatan Pengangkutan Jalan Malaysia (JPJ)	The Road Transport Department of Malaysia, responsible for vehicle registration and licensing.
Lembaga Hasil Dalam Negeri (LHDN)	The Inland Revenue Board of Malaysia, responsible for tax administration and compliance.
Person In Charge (PIC)	The staff member assigned to handle specific aspects of a booking (booking creation, key handover, inspection, or return).
Standard Operating Procedure (SOP)	Documented step-by-step instructions for performing routine operations consistently.
Surcharge	Additional fees applied for pickups or returns at non-standard locations or during specific time periods.
Work Breakdown Structure (WBS)	A hierarchical decomposition of project work into smaller, manageable tasks and deliverables.

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